

Chemistry Imp Topics

(in short)

Water – Hardness, softening, treatment,

Fuel – Types, calorific value, petroleum related topics

Corrosion – Types, prevention, electrochemical theory,

Industrial Materials

Cement – Composition, setting, manufacturing

Glass – Types, manufacturing, uses

Lubricants – Properties, types, mechanisms

Reaction Mechanism – Basic reaction types, intermediates

For 10 marks question Unit 1 -

* lime soda

* zeolite method

* ion exchange method

Unit 3

Mechanism of wet and dry corrosion and protection method

Short note on caustic embrittlement , galvanic cell and concentration corrosion, pilling bedworth rules , Stress corrosion .

Unit 4

Cement and glass

Chemical components and mechanism and application

Unit 5 SN1 and SN2 reaction

Elimination reaction mechanism

Unit 1 water Common impurities

Hardness of water and it's type

Degree of hardness

EDTA And EBT structure

Determination of hardness by EDTA method

Potable or drinking water

Disinfection

Filtration

*Break point chlorination

Sludge and scale

Casting imbrittlement

For 10 marks

* Lime soda method

* Zeolite or permutit method

* Ion exchange method

Unit 2 fuel Definition - fuel , calorific value , gross calorific value, net calorific value , carbonation

Classification of fuel and coal

Method of ultimate and proximate analysis

Dulong formula

Refining

Sweetening of petroleum

* frictional distillation

* Cracking

* octane number

* Cetane number

Unit 3 corrosion Definition

Effect

Types- dry corrosion

Wet corrosion

Acid formation

Briefly

Pulling bedworth rule

Passive metal

Protection method

Unit 4 Cement, glass

Chemical components

Mechanism and application

Definition of viscosity and viscosity index

Lubricant.

Unit 5 Structure of Aspirin and paracetamol, application

Markovnikov and anti Markovnikov rule, Saytzeff rule

Mechanism of substitution and elimination reaction

Difference between SN1 and SN2